Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Carter, Daniel C. Ph.D.		
Project Title:	Protein Crystal Growth Facility-Based Microgravity Hardware: Science and Applications		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Protein crystal growth		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35824	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-02
Start Date:	03/01/1999	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NAS8-99073		
Performance Goal No.:	1.21		
Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.		

Task Description:	Previously, in response to the first NASA Biotechnology NRA, we proposed to develop and construct a facility based protein crystal growth concept. This was undertaken largely because of the many important examples of quality improvements gained from growth in the diffusion-limited environment in space. The concept was based on the adaptation for microgravity of a commonly used disposable crystallization tray thereby improving sample logistics and handling, as well as greatly increasing the potential co-investigator group and science return. The facility for the expanded production of protein crystals for application in the structure determination is known as PCAM (Protein Crystallization Apparatus for Microgravity). The Diffusion-controlled Crystallization Apparatus for Microgravity (DCAM) was developed to take advantage of long duration opportunities available on Mir and the International Space Station. Access to the facilities is open to scientists from industry, academia and government laboratories. The principal aims of the research are: (1) Establish a protein crystal growth facility which would provide greatly increased experiment and co-investigator capacity in order to increase the odds of obtaining suitable crystals, consequently increasing the overall science return from each mission. (2) Produce protein crystals of improved size and order for in support of numerous structural biology and structure-based drug design research programs. (3) Utilize the facility to delineate factors contributing to the effect of microgravity on the growth and quality of protein crystals. The major objectives of this research are aimed at the production of protein crystals with improved size and perfection in support of numerous academic and industrial research programs in structural molecular biology. Improvements in the quality of protein crystals can be the limiting step in the determination of the atomic structure. A detailed knowledge of the atomic three-dimensional structure of protein molecules is key to a fundamental understanding of biochemistry and has important application in structure-based drug design. Further, examination of crystals grown in microgravity provides important scientific data which illuminates the mechanisms involved in the growth of high quality crystals in the microgravity environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 09/11/2009)